CERN Bulletin

Issue No. 47-48/2023 – 23 November 2023
http://home.cern/cern-people

The CMS collaboration at CERN presents its latest search for new exotic particles

This search for exotic long-lived particles looks at the possibility of “dark photon” production, which would occur when a Higgs boson decays into muons displaced in the detector

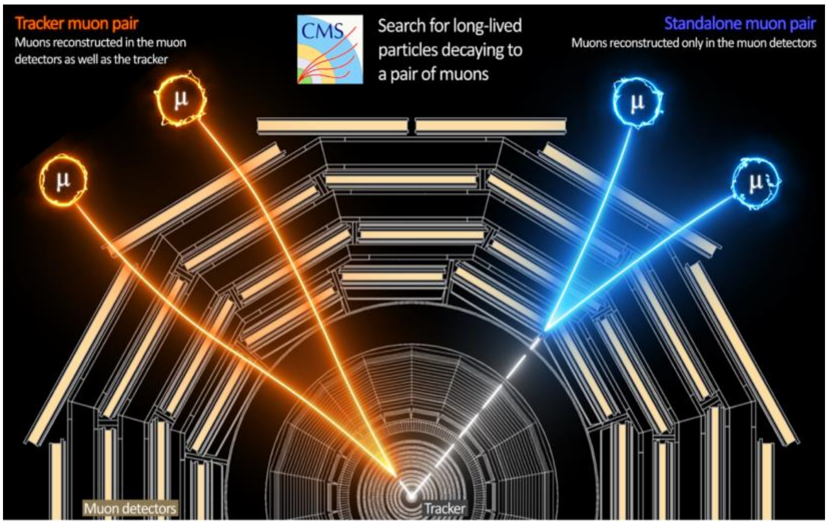


Illustration of two types of long-lived particles decaying into a pair of muons, showing how the signals of the muons can be traced back to the long-lived particle decay point using data from the tracker and muon detectors. (Image: CMS/CERN)

The CMS experiment has presented its first search for new physics using data from Run 3 of the Large Hadron Collider. The new study looks at the possibility of “dark photon” production in the decay of Higgs bosons in the detector. Dark photons are exotic long-lived particles: “long-lived” because they have an average lifetime of more than a tenth of a billionth of a second – a very long lifetime in terms of particles produced in the LHC – and “exotic” because they are not part of the Standard Model of particle physics. The Standard Model is the leading theory of the fundamental building blocks of the Universe, but many physics questions remain unanswered, and so searches for phenomena beyond the Standard Model continue. CMS’s new result defines more constrained limits on the parameters of the decay of Higgs bosons to dark photons, further narrowing down the area in which physicists can search for them.

In theory, dark photons would travel a measurable distance in the CMS detector before they decay into “displaced muons”. If scientists were to retrace the tracks of these muons, they would find that they don’t reach all the way to the collision point, because the tracks come from a particle that has already moved some distance away, without any trace.

Contents

Run 3 of the LHC began in July 2022 and has a higher instantaneous luminosity than previous LHC runs, meaning there are more collisions happening at any one moment for researchers to analyse. The LHC produces tens of millions of collisions every second, but only a few thousand of them can be stored, as recording every collision would quickly consume all the available data storage. This is why CMS is equipped with a real-time data selection algorithm called the trigger, which decides whether or not a given collision is interesting. Therefore, it is not only a higher volume of data that could help to reveal evidence of the dark photon, but also the way in which the trigger system is fine-tuned to look for specific phenomena.

"We have really improved our ability to trigger on displaced muons," says Juliette Alimena from the CMS experiment. "This allows us to collect much more events than before with muons that are displaced from the collision point by distances from a few hundred micrometres to several metres. Thanks to these improvements, if dark photons exist, CMS is now much more likely to find them."

The CMS trigger system has been crucial to this search, and was especially refined between Runs 2 and 3 to search for exotic long-lived particles. As a result, the collaboration has been able to use the LHC more efficiently, obtaining a strong result using just a third of the amount of data as previous searches. To do this, the CMS team refined the trigger system by adding a new algorithm called a non-pointing muon algorithm. This improvement meant that even with just four to five months of data from Run 3 in 2022, more displaced-muon events were recorded than in the much larger 2016–18 Run 2 dataset. The new coverage of the triggers vastly increases the momentum ranges of the muons that are picked up, allowing the team to explore new regions where long-lived particles may be hiding.

The CMS team will continue using the most powerful techniques to analyse all data taken in the remaining years of Run 3 operations, with the aim of further exploring physics beyond the Standard Model.

Host State Presidents visit CERN

President of the French Republic Emmanuel Macron during his state visit to Switzerland at the invitation of the President of the Swiss Confederation Alain Berset, Thursday 16 November 2023. (Image: CERN)

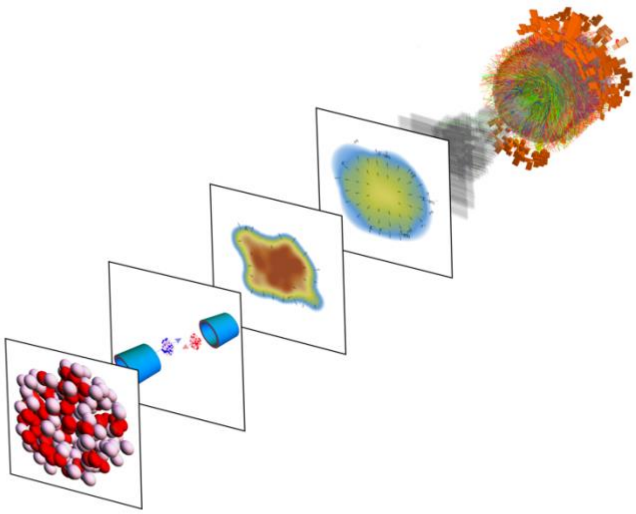
President of the French Republic Emmanuel Macron in the LHC tunnel on the occasion of his state visit to Switzerland, with the President of the Swiss Confederation Alain Berset and CERN Director-General Fabiola Gianotti. Speaking to journalists during the visit, President Macron said: "If I came here today, it is to reiterate my confidence in the scientific community and our ambition to maintain our leadership in this domain [particle physics]" (translated).

Original: « *Si j'ai voulu venir là aujourd'hui, c'est pour témoigner ma confiance aux équipes et notre volonté, notre ambition, de conserver la première place dans ce domaine [physique des particules].* »

Watch highlights from President Macron's visit to CERN here: https://videos.cern.ch/record/2299307

Thick-skinned: Scientists determine the thickness of neutron "skin" in lead-208 nuclei

This is the first measurement of the neutron skin of lead-208 using the strong force as a probe and it can provide insight into the structure of nuclei and neutron stars



When lead nuclei (left) are collided, the neutron distribution affects the shape of the quark–gluon plasma matter produced (middle), leaving measurable imprints in the distributions of detected particles (right). (Image: CERN)

Lead-208 has an intriguing nucleus. It is neutron rich, containing 82 protons and 126 neutrons. One of its more interesting properties is its structure: its centre is composed of both protons and neutrons, but at its edge, there is a diffuse shell of mostly neutrons. Scientists call this the neutron "skin." Research into the neutron skin can help deepen our understanding of quantum chromodynamics, that is, how quarks and gluons (exchange particles of the strong force) in the nucleus behave. This can also have applications in astrophysics, giving insight into the structure of neutron stars – the very dense core left behind after a star between 10 and 25 times the mass of the Sun has undergone a supernova explosion.

Using data from heavy-ion runs of the Large Hadron Collider, theoretical physicists at CERN have been able to determine the thickness of the lead-208 neutron skin to be 0.217±0.058 femtometres. This is in line with previous measurements obtained by other collaborations using different methods. In total, the scientists used 670 data points taken from Runs 1 and 2 of the LHC, mostly from the ALICE experiment, with some from ATLAS and CMS.

Measuring the thickness of this neutron skin is no easy feat. While the structure of the protons in the nucleus can be determined using electron scattering, neutrons have no charge, therefore they do not scatter electrons in the same way. However, neutrons are heavily affected by the strong nuclear force, which binds quarks and gluons together in the nuclei of atoms.

In the heavy-ion run of the Large Hadron Collider, beams of lead-208 are fired in opposite directions and made to collide at high energies. These nuclei are Lorentz-contracted by factors of up to 2500, meaning they are "squashed" into a flat pancake moving at close to the speed of light. At the moment of collision, because of the immense energy and pressure in the colliding nuclei, the gluons holding the quarks inside the nucleons together are ripped apart, creating a substance called quark–gluon plasma. Quark–gluon plasma is thought to be what the Universe consisted of moments after the Big Bang, and it is also theorised to be at the core of neutron stars.

As the pressure and temperature lower in the LHC, this plasma decays into particles, which can be tracked by LHC detectors to show the properties of the quark–gluon plasma. In lead-208, the distribution of the protons and neutrons determines the size and shape of the quark–gluon plasma. This allows physicists to "see" the lead-208 nucleus' structure and thus to calculate the thickness of its neutron skin.

This is the first time the thickness of the lead-208 neutron skin has been measured using the strong force. The Lead Radius Experiment (PREX) collaboration at the Thomas Jefferson National Accelerator Facility in Virginia obtained a similar result of 0.283±0.071 fm in 2021. However, this was determined using techniques using the electroweak force, as opposed to the strong force.

"What is exciting about this new determination is that it is done using only existing data and yet gives a competitive uncertainty when compared to other experimental determinations," said Wilke van der Schee, from CERN's Theoretical Physics department, one of the paper's authors. "In the future, more dedicated measurements can definitely improve the precision of the extraction from LHC data."

"The result connects the research of physicists from seemingly distant areas in an unprecedented way," adds Giuliano Giacalone, one of the paper's authors from Heidelberg University. "It brings together the topics of high-energy nuclear collisions, neutron stars and nuclear structure."

Naomi Dinmore

Progress Pride flag lights up the Esplanade des Particules in honour of LGBTQ+ STEM day

For the first time in its history, CERN raises the Progress Pride flag on the main Esplanade, showing solidarity with the worldwide scientific LGBTQ+ community

The "Progress Pride" flag at CERN, in full view of CERN personnel, local commuters and visitors to Science Gateway, cementing the LGBTQ+ community as a core part of CERN's research and collaboration. (Image: CERN)

On 17 November, at 12.45 p.m., a large crowd gathered on CERN's Esplanade des Particules to watch the Progress Pride flag rise at the Laboratory for the first time. The flag was raised to show CERN's commitment to recognising the talent and contributions that the LGBTQ+ community has made to scientific progress, both at CERN and around the world. The flag-raising ceremony coincides with LGBTQ+ STEM day, which was globally celebrated by scientific institutions on Saturday, 18 November.

"Today's raising of the Progress Pride flag is a very welcome symbol of the progress being made in acknowledging the presence and contribution of the LGBTQ+ community, not just at CERN but in the science and technology community worldwide," says Laura Stewart, from the CERN Cryogenics group. "We should not forget, however, that LGBTQ+ individuals have always been a part of that community, even if they have so often felt unacknowledged and invisible. I hope today's symbolic flag raising will finally close the chapter where many feel that they have to hide their true identities and open a new one where they can be free to contribute to their fullest, in the knowledge that they will be accepted for who they truly are and the excellence they bring to the world."

LGBTQ+ people in science have historically faced discrimination or have felt the need to hide their identity at work. A study from as recently as 2019 found that 28% of LGBTQ+ people working in the physical sciences have at some point considered leaving their job because of hostile attitudes towards their LGBTQ+ identity in their work environment. This is a loss for science, since, by nature, scientific research needs a diverse range of perspectives for effective collaboration and progress to take place. At CERN, which prides itself on its values of open collaboration in a melting pot of different cultures and backgrounds, it is therefore vital to recognise and celebrate diversity and ensure that the Organization is a welcoming place for people from all walks of life.

"The CERN community is full of so many diverse people and cultures, and their unique perspectives make the science that we do so much better," says Daniel Ally, from the CMS collaboration. "I'm glad to see CERN stand in solidarity with its LGBTQ+ community."

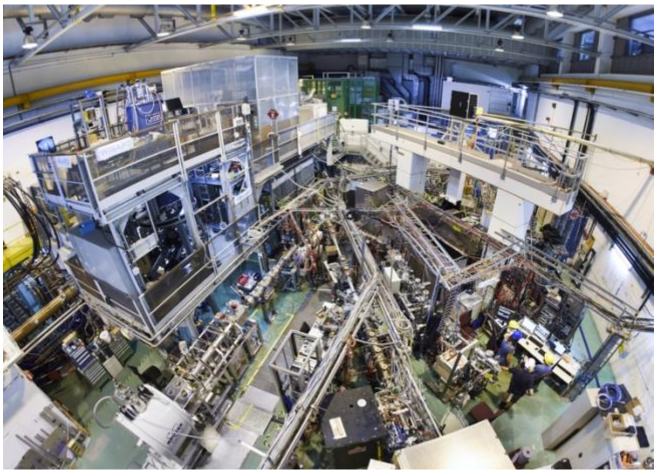
The flag was raised at 1.00 p.m. CET and was streamed live on social media. On the day, volunteers from the informal CERN LGBTQ+ network guided visitors around CERN's brand new Science Gateway facilities.

“Today we raise the flag as a sign of progress and pride at CERN,” says Louise Carvalho, CERN Diversity and Inclusion Programme leader. “We extend our hand to colleagues who hide their LGBTQ+ identity for fear of reprisal, exclusion or being passed over for promotion. Science is for everyone, and everyone belongs here – equally.”

Naomi Dinmore

ISOLDE sees shape shifting in gold nuclei

The finding comes a little more than 50 years after the phenomenon was first discovered at the facility in mercury nuclei



The ISOLDE facility hall, where the RILIS, Windmill and ISOLTRAP set-ups are located. (Image: CERN)

A little like humans, the nuclei of atoms tend to shrink as they lose weight. But atomic nuclei are complex quantum systems formed from neutrons and protons that are themselves composite particles made of quarks. As such, their usually spherical or nearly spherical shapes do not always simply shrink as particles are removed from their interior. In fact, exotic, neutron-deficient mercury and bismuth nuclei have been seen to alternate dramatically from football (soccer) to rugby ball shapes as single neutrons are removed from the nucleus.

A new study conducted at CERN’s ISOLDE facility now finds that neutron-deficient gold nuclei also display such shape shifting. The finding, described in a paper just published in Physical Review Letters, comes a little more than 50 years after the phenomenon was first discovered in the light mercury nuclei, also at ISOLDE. Belated happy golden anniversary, shape shifting.

To investigate gold nuclei, which have 79 protons, the team behind the new study used three set-ups at ISOLDE (called RILIS, Windmill and ISOLTRAP), each with a specific function. The combined power of the three set-ups allowed the team to determine the radii of gold nuclei with 104 neutrons all the way to 97 neutrons – that’s 21 fewer neutrons than that of the gold that can be found in rivers.

The results are striking. Given a previously observed transition from nearly spherical shape at neutron number 108 to a rugby ball at neutron number 107, followed by a plateau of rugby balls until neutron number 104, a possible continuation of the plateau and a transition back to nearly spherical shapes was expected. This continuation of the plateau was observed, as was the transition, which occurs at neutron number 101 and is a near-mirror image – centred at neutron number 104 – of the previously seen transition (see figure below). However, an extra shape-staggering blip was also observed at neutron number 99.

“This pattern in nuclear shape evolution is unlike any other we’ve seen before,” says James Cubiss, lead author of the paper. “In contrast to the mercury and bismuth nuclei, in which the staggering shows that football and rugby ball shapes are almost evenly matched in terms of the competition to minimise their binding energies, rugby balls win in the gold nuclei near neutron number 104. But they only just win, as evidenced by the shape staggering at neutron number 99 and the sudden jump back towards football-like shapes.”

The theorists in the team went on to perform state-of-the-art model calculations of nuclear radii and compare them with the data. They found that the model is able to follow the wandering gold

nuclei, but only if certain other data are used to constrain the calculations.
Fifty years on, the phenomenon of shape shifting still poses a tremendous challenge for nuclear theory.

Ana Lopes

CERN and the World Food Programme sign a memorandum of cooperation

The collaboration between the two international organisations will promote access to and use of breakthrough innovations in the field of IT to fight hunger worldwide

Enrica Porcari (left), Head of CERN's Information Technology Department and Dominik Heinrich, Director of Innovation, Change, and Knowledge Management at WFP, sign the Memorandum of Cooperation in Rome. (Image: CERN)

On 10 November, CERN and the World Food Programme (WFP) took a significant step in formalising their partnership by signing a memorandum of cooperation (MoC). This MoC paves the way for joint initiatives in which CERN's cutting-edge technologies – including artificial intelligence and quantum computing – will be made available to support WFP's worldwide efforts in the fight against hunger.
Today more than ever, our understanding of how hunger develops and evolves relies on technological innovations, in particular to analyse the large amounts of data needed to monitor situations and prepare humanitarian responses. Having access to the most appropriate computing technologies can make a huge difference and, ultimately, lead to a more efficient and accurate decision-making process, enabling early actions to be taken that save lives.
"We are grateful for CERN's collaborative spirit and forward-thinking vision, which enable us to harness the potential of technology and innovation in our ongoing battle against hunger," says Dominik Heinrich, Director of Innovation, Change and Knowledge Management at WFP. "WFP has always embraced innovation and partnerships like the one with CERN to further empower our mission, ultimately benefiting millions of people worldwide."
The areas that could benefit from the new cooperation framework are broad. They span food and drought monitoring, analysis and visualisation, and innovative Earth observation techniques for crisis indicators and early warning signals.
"WFP has, for a long time, been investing in ways to improve its work by embracing innovative technologies and processes and I am happy that our expertise will contribute to this effort," says Enrica Porcari, Head of CERN's IT department. "I am confident that by making premier technological innovations available to WFP, CERN will effectively contribute to saving and changing the lives of so many people worldwide."

Antonella del Rosso

CERN energy management: a collective effort

Find out how we can all help to improve CERN’s energy performance

CERN is committed to environmentally responsible research, and continual improvement of energy management is one of the key pillars of its strategy to minimise its impact on the environment. The global energy crisis of 2022 increased everyone’s awareness of the importance of effective energy management. It prompted the Organization, as a mark of its social responsibility, to take further measures to minimise its energy consumption. Now, we all have a role to play.

CERN has developed an Energy Policy that aligns with the ISO 50001 international standard used to continually monitor and improve energy management. The associated certification was awarded to the Organization on 2 February 2023, and yearly audits will be undertaken to check that CERN’s energy performance is improving. CERN’s commitments include keeping energy requirements to a minimum, improving energy efficiency and recovering waste energy. Measures taken in 2023 have included delaying the switch-on of heating, advancing the start of the annual year-end technical stop (YETS) and using the “eco-mode” for the LHC cryogenic installations whenever possible. In parallel, progress has been made on the approved heat recovery projects, aiming at reusing the waste heat of some of the LHC’s cooling towers and the new Data Centre in Prévessin, which has energy efficiency at its core. Finally, CERN is liaising with renewable electricity providers to try to secure part of its consumption volume (10% during run periods, 25% during long shutdowns) through private long-term contracts.

Practical tips for us all

- Every little helps: switch off lights and equipment when not in use, dress warmly instead of turning up the heating and, while ensuring regular ventilation for clean air, keep windows closed when possible.
- Find out more about CERN’s energy consumption via our Web Energy tool (https://energy.app.cern.ch/), which includes data from different sites, accelerators and experiments for 2023 and also for previous years.
- You will have seen awareness-raising posters across the CERN sites. The full set of posters is available here (https://edms.cern.ch/ui/file/2803716/Last_Released/EnergyMeasures_ALLPosters_Graphics.pdf) for you to help to spread the word.
- Familiarise yourself with CERN’s power cut response and what you should do in the event of a blackout (https://home.cern/news/official-news/cern/preparing-unlikely-event-blackout-cern).
- Share your feedback on energy saving via Mattermost or by emailing environment.info@cern.ch

For more information on energy management at CERN, visit https://environment.cern/content/energy-management.

HSE Unit

“Futur en tous genres” 2023 at CERN

On Thursday, 9 November 2023, CERN once again took part in the Futur en tous genres event. The aim of this initiative is to make schoolchildren aware of the diverse range of careers available to them and to encourage them to pursue their ambitions with confidence, free from gender stereotypes.

This year, 18 children aged 12 to 13 from Swiss schools had the opportunity to spend half a day in the company of CERN scientists. The programme included a visit to the cryogenics laboratory, a workshop on radiation protection, an introduction to jobs involving superconductivity and a guided tour of the ATLAS visitor centre. This half day, full of shared experiences, undoubtedly sparked the pupils' curiosity and, who knows, perhaps even paved the way for some future careers!

Mélissa Samson

Computer Security: Your six digits are yours

Recently, several colleagues (and perhaps even you?) have received WhatsApp messages that read something like “I’m sorry I sent you a 6-digit code by SMS by mistake, can you please pass it on to me? It’s urgent”, followed by an SMS with a code. Even though the message may be nicely disguised and look as if it comes from a trusted contact, even using comforting emojis (while putting pressure on you with “urgent!”), it can put your digital life on the line: depending on your response, everything could go down the drain…

…because this six-digit number is your account recovery code. It’s usually sent if you have lost access to your WhatsApp account. But this message was fake. It’s social engineering at its best and is sent by an attacker aiming to obtain your recovery code. They also do this to take over your WhatsApp account, and subsequently send similar malicious messages to your contacts, avalanching their attack. Your digital life could go down the drain* and, eventually, so too could that of your friends.

Don’t fall for it. As with passwords, your six digits are yours and yours only. This is true for WhatsApp codes, Instagram codes, CERN codes, passwords, PIN numbers and the six digits of your two-factor authenticator app or token. None of these digits should be shared. Ever. Not with your colleagues; not with your supervisor; not with the Service Desk; not with the Computer Security team (we have procedures in place to recover access to your CERN account if need be); not with anybody. Your password/PIN is yours and yours only. Sharing your six digits could negatively impact your private life (some colleagues who fell for the above scam are now in danger), your work and the overall reputation and operations of the Organization. This is a risk that you can only counter through vigilance and remembering that your six digits are yours and yours only.

** If you have already shared such a code, please follow this advice from WhatsApp ASAP (https://faq.whatsapp.com/1131652977717250). The Computer Security team cannot help you further with WhatsApp incidents as our official and supported method of communication is Mattermost, the reason being that CERN controls its security and privacy.*

Computer Security team

Official News

CERN Safety Rules: “Fire Safety and Radiation Resistance requirements for Cables”

The CERN Safety Rule listed below has been published on the CERN website dedicated to the Safety Rules:

Safety Guideline SG-FS-2-1-1 (revised version)

The Guideline complements the specific Safety Instruction SSI-FS-2-1 “Fire Safety and Radiation Resistance requirements for Cables” published in December 2021. It provides guidance as to the implementation of SSI-FS-2-1 in particular with regard to the certification and selection of Cable materials. The Guideline has been updated to define the conditions under which the HSE Unit accepts exceptions to Safety requirements with respect to fire performance of certain types of Cables.

The CERN Safety Rules apply to all persons under the Director General’s authority. They are available under the following link: http://www.cern.ch/safety-rules.

Tunnel linking the CERN sites: notice required in the event of exceptional opening

Use of the tunnel linking various parts of the CERN site is a special facility granted to the Organization by its Host States, France and Switzerland, to assist in its operations. The general principles for the use of the tunnel are laid down in agreements concluded between the Host States and between them and CERN (cf. Internal Rules CERN/DSU-DO/RH/8200). Failure to observe these rules may result in disciplinary action.

The tunnel is open **from 7.30 a.m. to 6.00 p.m., Monday to Friday**, excluding CERN's public holidays.

Outside the normal opening hours, members of the CERN personnel may request that the tunnel be opened exceptionally for the **urgent transport** of property belonging to CERN or the collaborating institutes. The request must be made to the Fire and Rescue service (tel. 74444) **at least two hours before** the intended time of passage through the tunnel to allow the Host State authorities to make any checks.

As with Gate E, people, goods and equipment passing through the tunnel may be subject to checks by the relevant French or Swiss authorities at any time. Members of the personnel are therefore reminded that, in addition to the transfer slip, if applicable, they must carry with them a travel document and their valid Swiss and French legitimation cards. Those who do not hold legitimation cards issued by the Host States due to the short duration of their contract with the Organization must carry with them the following documents:

- their national identity card, if accepted by the French and Swiss regulations, or a passport (with visa(s) if required by the French and/or Swiss regulations) and
- their blue CERN access card.

Host State Relations service
http://www.cern.ch/relations/
Tel. 75152

Fraudulent telephone calls: warning from the Swiss authorities

The Swiss Mission has informed CERN that the Geneva police force regularly receives reports of fraudulent telephone calls from scammers posing as bank employees, police officers or representatives of the authorities. The Geneva police force warns the public to beware of such scams and advises anyone receiving a seemingly fraudulent telephone call to hang up immediately and avoid communicating any personal or banking details. More information on what to do if you are contacted by a scammer is available on the following web pages:

- https://www.ge.ch/actualite/faux-employe-banque-nouvelles-escroqueries-telephoniques-aupres-seniors-4-05-2023 (in French)
- https://www.bazg.admin.ch/bazg/en/home/teaser-homepage/focus-teaser/warnung-telefonanrufe-mit-automatisierter-ansage-des-zolls.html

In addition, the Swiss Mission draws our attention to the website of the National Cyber Security Centre (NCSC), which lists all known cyber threats and provides practical information about fraud risks. Incidents can be reported on the website: https://www.ncsc.admin.ch/ncsc/en/home.html.

Host State Relations service
http://www.cern.ch/relations/
Tel. 75152

Announcements

28 November | Next blood donation campaign

Contribute to this essential act of solidarity that benefits us all

The CERN community is warmly invited to At the last blood donation campaign on 17 July 2023, 120 donors came along to give blood, of which 34 were first-timers. 76 donations were collected. Before the collection, the HUG presented a shortage of group A+ blood. Thanks to this collection, we were able to increase the blood reserves of group A+. This is a great success and we thank you for your contribution!

Nevertheless, the Geneva blood barometer, which is regularly updated, shows that blood stocks continue to run extremely low. CERN will hold a third blood donation session on **Tuesday 28 November, from 8.30 a.m. to 4.30 p.m.** in collaboration with the Hôpitaux universitaires de Genève (HUG), in Building 29.

Before coming to the **Building 29 donation point**, please make sure that:

- you are in good health and have no symptoms such as fever, cough, a cold or breathing difficulties;
- you are eligible to give blood:
    - Complete the brief online pre-donation questionnaire (in French)
    - please consult this information sheet and complete this questionnaire issued by the HUG (but note that only the pre-donation conversation with the nurse or doctor on the day can confirm your eligibility).
    - If returning from travel abroad or arriving from another country, perform the travelcheck on the HUG website (in French)
    - If you have recently been vaccinated against the flu or COVID,

ensure 48 hours have passed since receiving the vaccination.

***New*: possibility to make an appointment in advance**

Based on the experience of the donation campaign in July, the possibility of pre-booking your appointment via the OneDoc platform is made available on this link (https://www.onedoc.ch/fr/widget/8abfe0a6b86b907ca52e2fc093f1134a01b8b1b2d7eddc38e38d099bcfece834/book). Spontaneous donors will be able to come at their convenience without an appointment, mindful that there may be some waiting time on the day.

As a thank you, the HUG will be giving each donor a 10 CHF voucher, to be used in Novae's Restaurants 1 and 2 at CERN.

We hope to see many of you at the donation point!

---

30 November | CERN screening of "The Geneva Event"

Journey back 40 years to relive the excitement of hunting for the W boson at CERN, with an exclusive screening of the 1983 BBC documentary *The Geneva Event* on **Thursday, 30 November** from **1.00 p.m.** in the **Main Auditorium** https://indico.cern.ch/e/geneva-event.

The BBC Horizon team captured Carlo Rubbia in his element as it followed physicists and engineers at the SPS and the UA1 and UA2 experiments from 1978 to 1983 on the path to one of the most important discoveries in particle physics. Catch a glimpse of old control rooms, the expanding CERN sites and newly installed state-of-the art technology. The documentary also features the first antiproton beams, retro animations, lectures by Steven Weinberg and Abdus Salam, as well as former UK Prime Minister Margaret Thatcher's visit to CERN.

Scientifically robust, this classic documentary film offers a rare insight into the technological challenges and personal stories behind large physics experiments.

---

30 November | CERN Colloquium | The Einstein Telescope: the next-generation gravitational-wave observatory in Europe

The Einstein Telescope: the next-generation gravitational-wave observatory in Europe, by Dr Michele Punturo (INFN, Perugia).

Thursday, 30 November 2023, 4.30 p.m.–5.30 p.m.

Room 222/R-001

Live webcast available at: https://webcast.web.cern.ch/event/i1329008

Gravitational waves (GWs) are the newest tool for exploring the Universe. Advanced Virgo and Advanced LIGO have opened a new window into the Universe by detecting GW signals in the Hz–kHz frequency range. Pulsar timing array experiments have just announced the detection of GWs in the nanohertz frequency range.

A new generation of GW interferometric observatories is being prepared to replace the current generation of GW detectors within the next decade. This will make it possible to search almost the entire Universe for GW signals. The Einstein Telescope (ET) is at the forefront of the design, preparation and implementation of a next-generation gravitational-wave observatory in Europe. An overview of the observatory's scientific objectives, design, required technologies and project organisation will be presented at the event.

______

More information available on Indico: https://indico.cern.ch/event/1329008/.

30 November | Alumni event: News from the Lab with the CERN & Society Foundation

The purpose of the CERN & Society Foundation is to support and promote the mission of CERN and disseminate its benefits to the wider public. The Foundation operates nationally and internationally to pursue its mission, across three main areas: Education & Outreach, Innovation & Knowledge Exchange, and Culture & Creativity.
30 November – 6 p.m. on Zoom
More information and registration on the Alumni website: https://alumni.cern/events/127693

---

8 December | Music and physics: a space–time voyage back to our origins, with Yo-Yo Ma

Register now to join world-famous cellist Yo-Yo Ma for a unique evening of music and physics, with CERN Director-General Fabiola Gianotti

On Friday, 8 December, renowned cellist Yo-Yo Ma will be the exceptional guest of a special evening devoted to music and physics.
This unique in-person-only event will explore the mysteries, beauty and symmetry of both music and physics. It will take place in the CERN Science Gateway Sergio Marchionne Auditorium starting at 7.00 p.m.
A limited number of seats have been reserved for the CERN community. Please register here (https://indico.cern.ch/e/music-and-physics-CERN).

Become an ambassador for Women and Girls in Science and Technology!

From 5 to 9 February 2024, female science ambassadors are invited to share their passion for science with local schoolchildren. Volunteer to take part!

CERN For the eighth year running, CERN, the UNIGE Faculty of Science, EPFL and the Laboratoire d'Annecy de Physique des Particules (LAPP) are joining forces to celebrate the International Day of Women and Girls in Science. During the week of 5 to 9 February 2024, female scientists and engineers will be visiting local schools to inspire younger generations to explore the world of science.
As a female science ambassador, you will talk about your career path, share your projects and professional experiences and maybe even give a short demonstration. Our aim is to change the way schoolchildren view scientific, technical and technological professions, making them accessible to both girls and boys. And who knows, these presentations might even inspire some future careers!
Every year, the Women and Girls in Science and Technology week proves to be a great success. In 2023, over 240 presentations were given by 100 or so female science ambassadors to 5220 schoolchildren! **This is why we're always looking for more female scientists willing to give up a bit of their time to give talks in schools. So come and join the adventure by signing up: the deadline is 10 December 2023 (11.59 p.m.).**

Conditions of participation:

- Registration is open to all women working in a profession connected with science, technology, engineering or maths (STEM), as well as computer science, communication or education.
- Priority will be given to presenters from CERN, UNIGE, EPFL and LAPP. If you are not from one of these institutions but would like to take part, please contact us.
- Volunteers will deliver one-hour presentations for a maximum of 35 pupils aged 7 to 15.
- The majority (95%) of the presentations will be given in French, but presentations in English are also possible.
- Volunteers are required to attend a briefing session.

Sign up and find out more: http://cern.ch/fds-interne
Deadline for signing up: 10 December 2023 (11.59 p.m.)

For CERN volunteers only: we are also looking for a female scientist who is available on Saturday, 24 February at 10.30 a.m. to talk (in French) to children aged 7 and over about her job and about particle physics in general. This event is being organised by the Bourg-en-Bresse multimedia library. Transport expenses will be covered. For more information and to volunteer, contact education.locale@cern.ch.

Thank you for volunteering!

Visit CERN Science Gateway during the winter holidays

When the CERN site closes its doors on 23 December for the end-of-year closure, CERN Science Gateway will remain open for you to enjoy with your friends and family.
CERN Science Gateway exhibitions will be welcoming visitors for most of the winter break, from Tuesday to Sunday, 9.00 a.m. to 5.00 p.m., closing only on 24, 25 and 31 December and 1, 6 and 7 January.
Families and individuals can arrive on the day and book activities on the spot (https://visit.cern/families-individual-visitors) (groups of 12 or more need to book visits several months in advance). Explore the exhibitions, the Big Bang Café and the CERN shop. Note that during December CERN personnel receive a 10% discount at the CERN shop when showing their CERN access card.
Public transport is recommended, as only a limited number of paid parking spaces are available. For more information visit: https://visits.web.cern.ch/getting-here
CERN Science Gateway guides have welcomed more than 40 000 visitors since the centre opened to the public on 8 October. Guide training is still available if you would like to join the team.
Come and visit for yourselves and enjoy CERN Science Gateway.

Obituaries

Bikash Sinha (1945 – 2023)

Bikash Sinha, influential Indian scientist and pioneer in quark–gluon plasma and the early Universe, passed away on 11 August at the age of 78. As one of the ALICE experiment's early visionaries and architects, his impact on heavy-ion physics is unmistakable.
Bikash Sinha was born on 16 June 1945 in Kandi, Murshidabad, in the state of West Bengal, India.

After graduating from Presidency College, Kolkata, with a degree in physics in 1964, he went on to obtain the Tripos in Natural Sciences from King's College, Cambridge, in 1967 and then a PhD in nuclear physics from the University of London in 1970. He returned to India at the invitation of nuclear physicist Raja Ramanna and joined the Bhabha Atomic Research Centre (BARC) in 1976. In the early 1980s, he started working in high-energy physics, particularly in relativistic heavy-ion collisions and the formation of quark–gluon plasma. He was appointed Director of the Variable Energy Cyclotron Centre (VECC) in 1987 and was concurrently Director of the Saha Institute of Nuclear Physics (SINP) from 1992 to June 2009. His numerous awards and honours include the Padma Shri Award in 2001 and the prestigious Padma Bhusan Award (the third-highest civilian award in India) in 2010 for his significant contribution to science and technology. He was also a member of the Scientific Advisory Council to the Prime Minister of India.

Bikash Sinha at CERN.

As director of two major institutes in Kolkata, his efforts put India on the map of nuclear and particle physics laboratories. He was a strong supporter of India's engagement with the international scientific community via CERN's programmes. His charismatic leadership and charming personality allowed him to successfully navigate the scientific bureaucracy surrounding a multi-agency funding model for the nascent ALICE collaboration. From modest beginnings at the CERN SPS in the early 1990s, armed with only a handful of collaborators, students and borrowed equipment, but with a grand vision and unbeatable spirit, he nourished and led the Indian team to become a major pillar of ALICE and a key player in heavy-ion physics.

Bikash was a synthesis of science, culture, philosophy and society. He initiated the creation of a medical cyclotron in Kolkata to diagnose and treat prostate cancer. Inspired by the great Indian poet and Nobel laureate Rabindranath Tagore, he organised a one-of-a-kind international conference (https://cerncourier.com/a/a-carnival-of-ideas-in-kolkata/), MMAP (Microcosmos, Macrocosmos, Accelerator and Philosophy), in May 2022, combining elementary particles, the Universe, accelerator physics and philosophy through poetry and songs.

He initiated a successful series of international conferences on the physics and astrophysics of quark–gluon plasma (ICPAQGP) that have been held in India since 1988, and he organised and chaired the Quark Matter Conference in India in 2008. His efforts helped India to become a prominent CERN non-Member State, culminating in its accession to Associate Member State in 2016.

While Bikash's passing leaves an undeniable void, his legacy is a vibrant and thriving team, primed to continue his journey. We will always remember him for his charismatic personality, great kindness, openness and generosity. We honour his memory and, with our deepest condolences, extend our sympathy to his family.

His friends and colleagues in the ALICE collaboration